Subset Sums in a Fourfold Sumset

A sharper non-progression lemma and the audited bound $N \le 94$

Sum-of-Subset research project

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Abstract

We continue the independently audited chain from $N \le 95$ to $N \le 94$. The main methodological improvement precedes the endpoint computation: we prove $\rho(X) \ge |X| - 5$ for a positive real set with non-progression subset-sum support and $|X| \ge 17$. This strengthens the previously used $|X| - 14$ bound and lowers the archived sharp lemma's threshold from 21 to 17. The finite base has 2,723 deleted intervals, each independently checked by two exact support computations. At the size-95 endpoint it eliminates the entire positive-low-hole three-empty family and removes 10,280 low-hole root systems before geometric solving. Loss-dependent inequalities are selected by explicit group-wide threshold tests. The complete endpoint proof uses fresh integer certificates; no archived endpoint certificate is read. The stronger archived bound 84 remains outside the complete independent replay: this report advances the audited chain and does not surpass 84.

1 Statement, notation, and dependencies

For a finite real set A, let $\mathrm{FS}(A) = \{\sum_{x\in X} x : X \subseteq A\}$ include the empty sum and let $4A = A + A + A + A$ allow repeated summands. Let N be the largest feasible cardinality for $\mathrm{FS}(A) \subseteq 4A$.

Theorem 1.1. *Every feasible set has at most 94 elements. Thus* $71 \le N \le 94$.

The lower bound is the unchanged set $[-31, 31] \cap \mathbb{Z} \cup \{\pm 93, \pm 155, \pm 217, \pm 713\}$. We use the audited chain $115 \to 103 \to 100 \to 99 \to 98 \to 97 \to 96 \to 95$, ending in [1]. Its integralization, covering, peeling, small-increment inverse, simultaneous inverse, span, effective-hole, sharp four-extreme, mass, and pairing lemmas remain the foundation. The new defect argument develops the archived strategy [3], with a larger independently enumerated base and a smaller threshold.

A hypothetical size-95 example contains zero, since adjoining zero otherwise produces a forbidden size-96 example. No hereditary feasibility under arbitrary deletion is assumed. Integralize, remove four original outer elements on each side to obtain D, and peel twice more to $D \supset D_1 \supset E = P \cup \{0\} \cup (-Q)$. Put

$$d = 87, \quad e = 79, \quad m = a + b = 78, \quad a = |P|, \quad b = |Q|,$$
$$T_r = r(r+1)/2, \quad I = (a - 39)^2, \quad \Lambda = 1 + T_a + T_b = 1561 + I, \quad \Omega = 450 - I.$$

For a positive set X of size r, write

$$\mu(X) = |\,\mathrm{FS}(X)| - 1 - T_r, \quad \delta_0(X) = |\{x \in X : 2x \notin R(X \cup \{0\})\}|, \quad \rho(X) = \mu(X) + 2 - 2\delta_0(X),$$

where R denotes distinct-element two-sums. The inherited bounds from [2] are $\rho \ge -8$ in general, $\rho \ge 0$ for $r \ge 14$, and $\mu \ge r - 3$, $\rho \ge r - 14$ for a non-progression support in the ranges used below. For increasing prefixes X_i, the increments $y_i = |\,\mathrm{FS}(X_i)| - |\,\mathrm{FS}(X_{i-1})| - i$ are nonnegative integers, and

$$\rho(X_i) \ge \rho(X_{i-1}) + y_i - 2. \tag{1}$$

Indeed, μ increases by y_i and the number of diagonal defects increases by at most one. If $i \ge 6$ and $y_i \le i - 5$, the inherited small-increment inverse expresses X_i, up to positive scaling, as $[1, i + q]_{\mathbb{Z}} \setminus H$, where $q \le y_i$ and the maximum is retained.

2 A sharp defect bound starting at size 17

Theorem 2.1. *Let X be a positive real set of size $r \geq 17$. If $\mathrm{FS}(X)$ is not an arithmetic progression, then $\rho(X) \geq r - 5$. The constant 5 is sharp.*

We first prove the deleted-interval input, including its complete finite base. The computer checks in this section form an explicit finite part of the proof; the following reduction supplies the unbounded part.

Lemma 2.2. *If $Y = [1, j+q]_{\mathbb{Z}} \setminus H$, $j \geq 8$, $0 \leq q \leq 3$, $|H| = q$, and $j+q \notin H$, then a non-progression $\mathrm{FS}(Y)$ satisfies $\rho(Y) \geq j - 5$.*

Proof. The density forces $\gcd(Y) = 1$, so an arithmetic-progression support is exactly a full interval. The case $q = 0$ is full. For $j \geq 10$, $j + q \leq 3j/2 - 2$, and Lev's central-interval theorem [4, Theorem 1] gives

$$[2q + 1, \textstyle\sum Y - (2q + 1)]_{\mathbb{Z}} \subseteq \mathrm{FS}(Y).$$

By subset-complement symmetry, the total number of holes is twice the number of missing sums in $[1, 2q]$. These small sums depend only on which positions through $2q$ are deleted.

Every diagonal defect lies in $[1, q]$ or $[j, j+q]$. For an interior x, the $\min(x, j+q-x) > q$ distinct pairs centered at x in $[0, j+q]$ are disjoint, so q deletions cannot destroy them all. Low defects depend only on deleted positions through $2q$; high defects depend only on deleted depths through $2q$ below the maximum.

For $j \geq 14$, preserve these low positions and high depths, with respective counts l, u. Replace the other $t = q - l - u$ deleted depths by $2q + 1, \ldots, 2q + t$ in an interval with 14 retained elements. These positions fit between the two endpoint bands for every $q \leq 3$. The resulting set Y_0 has the same number of support holes and the same number of diagonal defects. With $h = j + q$, the exact identity is

$$\mu(Y) = \sum_{v \in H} (h - v) - T_{q-1} - \mathrm{holes}(\mathrm{FS}(Y)).$$

Each low deletion contributes $j - 14$ more to this depth sum than in Y_0. High depths do not change, and the ordered middle depths are at least the replacement depths. Non-full support requires $l \geq 1$: otherwise every integer through $2q$ is present and there are no holes. Consequently

$$\rho(Y) - j \geq \rho(Y_0) - 14 + (l - 1)(j - 14) \geq \rho(Y_0) - 14.$$

It remains to check exactly $j = 8, \ldots, 14$, $q = 1, 2, 3$, and every q-subset of $[1, j + q - 1]$. The new checker `verification/nonap_sharp.py` computes subset sums both as a set of integers and as an integer bitset, compares the results, and forms all distinct-pair sums of $Y \cup \{0\}$ to count defects. The counts are

j	8	9	10	11	12	13	14
All deletion sets	164	219	285	363	454	559	679
Non-full supports	88	108	130	154	180	208	238
Minimum $\rho(Y) - j$	-5	-5	-5	-5	-5	-5	-5

All 2,723 cases, including all 1,106 non-full supports, pass. The checker also verifies the depth identity and defect bands in every base case. It checks the central-interval hole identity only for $j \geq 10$; that theorem is not assumed at sizes 8 or 9. The preceding signature reduction now proves the assertion for every $j \geq 8$. □

Proof of Theorem 2.1. If every $y_i \geq 4$ for $8 \leq i \leq r$, start at $\rho(X_7) \geq -8$ and use (1):

$$\rho(X_r) \geq -8 + 2(r - 7) = 2r - 22 \geq r - 5 \qquad (r \geq 17).$$

Otherwise take the last $j \geq 8$ with $y_j \leq 3$. Since $3 \leq j - 5$, the small-increment inverse applies. If $\mathrm{FS}(X_j)$ is not a progression, Lemma 2.2 gives $\rho(X_j) \geq j - 5$; all later increments increase ρ by

at least two, proving the result. If the support is a progression, let $k > j$ be the first index whose support is not. The preceding progression and its new translate must be disjoint: an intersection places the translating element on the same step and makes their union another progression. Hence

$$y_k = |\operatorname{FS}(X_{k-1})| - k \geq 1 + T_{k-1} - k, \qquad \rho(X_k) \geq T_{k-1} - k - 9 \geq k - 5$$

for $k \geq 9$. Apply the same later-increment argument. For sharpness, take $X = \{1, 3, 4, \ldots, r+1\}$. Lev's theorem with $q = 1$ shows that the only missing sums in its hull are 2 and $\sum X - 2$. Its only diagonal defects are 1 and $r + 1$; for 3 use $0 + 6$, and for other nonmaximal elements use their two neighbors. Thus $\mu = r - 3$ and $\rho = r - 5$. □

Compared with the archived threshold-21 proof, adding the 383 cases at sizes 8 and 9 permits the recurrence to start two steps earlier and lowers the threshold by four. We make no claim that 17 is the optimal threshold; it is the threshold proved and used here.

3 Lattice reduction without a circular use of the new budget

The nonnegative inherited identity is

$$\rho(P) + \rho(Q) + 2\vartheta + c + \mathcal{T} = \Omega, \quad c = 2|2D| + 18d + 85 - |\operatorname{FS}(D)|, \quad \mathcal{T} = \kappa + \ell_1 + \ell_2 + g, \tag{2}$$

where $\kappa = |\operatorname{FS}(E)| - \Lambda - \mu(P) - \mu(Q)$ and the other terms are the same nonnegative peeling and diagonal slacks as in [1]. A deficient sign would give $1 + T_{79} > 24 \cdot 95 + 77$. The paired bootstrap gives $q^2 \leq 2416$, with even q, hence $q \leq 48$ and $a, b \geq 15$. Nonnegative Ω then leaves $a = 18, \ldots, 60$. For all 43 oriented pairs, $T_{\min(a,b)} - 2 - \Omega \geq 68 > 0$. The simultaneous inverse theorem therefore supplies one common core lattice, with holes

$$h_P, h_Q \leq \kappa, \quad z = h_P + h_Q - \kappa, \quad 0 \leq z \leq \min(h_P, h_Q) \leq \kappa, \quad h_X < |\operatorname{FS}(X)| - 1.$$

This step has not assumed that D_1 or D is on that lattice.

The inherited universal span comparison, with $C = 2m + 4e + 14 - \Lambda = -1075 - I$, gives

$$\begin{aligned} |\operatorname{FS}(D_1)| - 2|\operatorname{FS}(E)| &\leq C + \Omega + \rho - \mu + 2z, \\ |\operatorname{FS}(D_1)| - 2|\operatorname{FS}(E)| &\leq C + 3\Omega - \rho - \mu, \end{aligned}$$

where ρ, μ denote the two side sums. The maximum of a positive set is defective, so $\rho \leq \mu$. If $z \leq 14$, the first bound is at most $-597 - 2I < 0$. If $z \geq 15$, both supports are non-progressions. Even the earlier bounds $\rho \geq 50$, $\mu \geq 72$ leave only $|a - 39| \leq 6$. These sides have size at least 33. The effective-hole witness $(r, C_0, \beta) = (33, 89, 7)$, checked with strict integer tests, gives $\mu \geq 78 + 180 = 258$. Thus $\rho + \mu \geq 308 > 275 - 4I$. The maximum resulting high-hole comparison is -33. The first-layer comparison has margin $\Lambda - (4d - 10 + \Omega) = 773 + 2I > 0$. An off-lattice element would give two disjoint translates of the smaller subset-sum support, contradicting the strict factor-two comparisons. Thus D_1 and then D are on the core lattice.

Only after this reduction do hole propagation and the sharp four-extreme theorem of [1] give

$$w \leq 4z, \qquad \mathcal{T} \geq \kappa + 164 - 3z, \qquad \mathcal{T} \geq \max(55, z, 164 - 2z). \tag{3}$$

The new defect theorem is not needed for these lattice comparisons; its use below is therefore independent of the sharper post-lattice budget. All core sides now have at least 18 elements, so its threshold 17 is always satisfied.

4 Scalar pruning and common endpoint models

Let j of the eighteen covering units be empty, and put $\delta = c - 87j$. The universal span estimate combined with (2)–(3) yields

$$R_p + R_q + \delta \le 368 - I - 87j + 4z.$$

For zero holes, $c \le 286 - I$. For $1 \le z \le 14$, both sides are non-progressions and Theorem 2.1 gives $\rho \ge (a-5) + (b-5) = 68$. Therefore

$$c \le 218 - I + 2z \le 246 - I < 3d.$$

In particular there are no positive-low-hole three-empty modes. The complete low-hole ranges and counts are

Regime	Z_0	Z_1	Z_2	Z_3	P_0	P_1	P_2
a	23–55	25–53	29–49	34–44	24–54	27–51	31–47
Integer roots	132	2030	12138	32912	124	1750	9826

Their total is 58,912. The weaker $\rho \ge 50$ would retain 69,192 low roots, including 8,976 positive three-empty roots. The sharper lemma removes those roots and 1,304 additional positive roots, a total reduction of 10,280 before any linear feasibility computation. The zero-hole three-empty family is retained in full.

For high holes the effective-hole search is the same bounded search as in [1], with every chosen parameter rechecked exactly. It gives lower bounds u_p, u_q for $\mu(P), \mu(Q)$. Replace the defect floor in the flexible inverse by the sharper floor:

$$r_*(r,u) = \max\{r-5, u+2-2r, L_2(r,u), \dots, L_{10}(r,u)\},$$

where, for $v = \max(0, u - 4\beta + 2) = (\beta+1)q + s$,

$$L_\beta(r,u) = \min\{(\beta-1)(r-14), (\beta-1)q + \min(s, \beta-1)\}.$$

The constant $r - 14$ inside this inherited flexible-inverse formula is not silently changed. Thus

$$c \le 450 - I - r_*(a,u_p) - r_*(b,u_q) - \max(55, z, 164 - 2z).$$

There are 3,500 surviving oriented scalar states and $c \le 171 < 174$, so at most one unit is empty. The active-center estimate uses the orientation fixed by the first pair's actual contributor, exactly as in [1]; selecting a stronger reflected orientation is not allowed. Against $F \le 1692 + 4z$ it leaves 959 states for $j = 0$. There are 1,371 states with enough cap for $j = 1$.

For each fixed (a, j, u_p, u_q), take the largest hole count Z and largest cap C_*. Replace the two mass coordinates by $\widehat{S}_p = S_p + Z - z$, $\widehat{S}_q = S_q + Z - z$. All mass, cardinality, span, loss and grouped-overlap rows retain their validity by the transport lemma of [1]; upper endpoint rows weaken. The 2,330 retained states merge into 50 groups: 15 of type H_0 and 35 of type H_1. This reduces high-hole integer models from $959 \cdot 4 + 1371 \cdot 70 = 99806$ to $15 \cdot 4 + 35 \cdot 70 = 2510$. For low positive holes the same transport uses $Z = 14$.

5 Threshold-driven rows and the unified covering budget

The twenty centers, eighteen paired units, actual-contributor choices, and the grouped union-deficit variables are exactly those proved in [1]. For $j = 0, 1, 2, 3$, their mode counts are $4, 70, 578, 2992$. All missing-unit subsets and both choices in a potentially split pair are retained. Let l_v, u_v be the hull losses at a retained center, and let L_p, L_q be the actual-target losses of the two leading slots. The common residual budget contains these losses and the nonnegative actual deficits of disjoint groups of slot labels.

For clarity, for k ordinary same-sign slots the core-hull lower bound on their joint deficit is

$$79k - (R_p + R_q + v_{\max} - v_{\min} + 1) + \max_v(l_v + u_v).$$

For a leading slot and k own-center slots the corresponding lower bound is

$$79(k+2) - 2 - p_{\max} - R_p - 2R_q + max(L_p, l_1 + u_1, \ldots, l_k + u_k).$$

Both also dominate the sum of individual losses. These follow by counting the common core hull and then the noncore points, and adding points of the raw union outside the actual target. Each whole-block variable is the actual deficit of that block, so alternative subset bounds constrain one variable; they are not separate overlap gains added repeatedly. Sum only deficits of disjoint blocks. The leading centers are integral with at most one empty unit by the inherited contributor lemma. With two or three empty units, $\delta \le 112 < 173 \le |2D|$, so both leading slots contribute and their centers are again integral. This justifies all core-hull counts.

5.1 Every conditional row has its own checked threshold

For each structural group g the checker derives its maximum residual cap C_g from its entire integer side-size interval, or from its high cap C_*. It also takes the smallest side size in that group. The following tests are independent:

Rows used	Required group-wide condition
Ordinary individual quantiles	$C_g < \min(a, b) + 5$ throughout the group
Core individual quantiles	$C_g < 83$
Basic leading quantiles	$C_g < 173$
Nested leading quantiles	$C_g < 165$
Strict outer order	$j \le 1$, or $j = 2$ with contributing leading slots

Use the strongest justified individual quantile type; otherwise use global quantiles. This avoids two invalid substitutions from the preceding endpoint: an H_1 cap can now be 84, so core quantiles are not automatic; an H_0 cap can be 171, below 173 but above 165, so basic leading rows do not automatically justify nested ones. All three-empty models retain weak outer order. Z_3 has residual cap at most 25 and uses ordinary quantiles; P_2 has cap at most 72 and uses core quantiles. The code computes these choices, rather than accepting a certificate's choice of applicability conditions.

During high-hole transport, freeze the weaker endpoint group's row policy when comparing a source state with its endpoint. A smaller source cap could justify additional rows, but those rows are not needed for the common endpoint model. This keeps the matrix and row labels fixed and proves coefficientwise transport dominance without conflating the two leading thresholds.

5.2 Numerical constants in the necessary models

Write $K = |2D|$, $x_0 = R_p < x_1 < x_2 < x_3 < x_4 = M_p$, and similarly y_i. The inherited mass and gap rows on each side are

$$R_p \ge a, \qquad M_p - R_p \ge 4, \qquad x_i - x_{i-1} \ge 1,$$
$$2M_p + 6R_p - 2S_p + 2z \le -2T_a - 12 - 2u_p,$$
$$M_p + 4R_p - S_p \le -T_{a-1} - 6,$$
$$x_1 + x_2 + x_3 + M_p - S_p + z \le -T_a - u_p,$$
$$R_p + x_1 + x_2 + x_3 + M_p - S_p \le -T_{a-1}.$$

Here $u_p = 0$ for Z, $a - 3$ for P, and the certified high-hole lower bound for H; all occurrences of z are transported together. The common cardinality row is

$$S_p + S_q + \delta - 2K - 4z \le 1650 - 87j.$$

At a positive center v the global quantiles are $v + l_v + M_p \geq 87$, $v - u_v - M_q \leq S_p - 86$. The core versions replace M_p, M_q by R_p, R_q and 87,86 by 83,82; the ordinary versions are $v + l_v \geq b + 5$, $v - u_v \leq S_p - a - 4$. Individual loss caps are 86 globally and 82 for core rows. Reflection supplies the negative side.

The basic leading rows are

$$c_p + 2M_p + L_p \geq K, \quad S_p + 2M_q - c_p + L_p \geq K - 1, \quad S_p + L_p \geq K.$$

Their coarse versions use $K \geq 173$. The nested rows, used only under their stronger cap test, are

$$c_p + 2x_i + L_p \geq 165 + 2i, \quad S_p + 2y_i - c_p + L_p \geq 164 + 2i \qquad (0 \leq i \leq 4).$$

They count the contributors from a subset of D of size $e + 4 + i$, whose twofold sumset has at least $2(e + 4 + i) - 1 = 165 + 2i$ elements. The cap below 165 ensures the counting interval is nonempty.

The coarse model substitutes $K \leq M_p + M_q + R_p + R_q + 21$ into the cardinality row, eliminating K and the inner coordinates. Difficult groups retain all five inherited nested interval-union envelopes, plus the 25 independent-cutoff rows

$$K \leq M_p + M_q + x_i + y_h + 1 + T_{4-i} + T_{4-h} \quad (0 \leq i, h \leq 4).$$

The latter count the integer central interval and all pairs among the remaining same-sign outliers. No new unproved twofold-sumset estimate is inserted.

5.3 Exhaustive local alternatives

The four types of binary alternatives from [1] remain valid, with current constants. For a retained center, either $l_v + u_v \geq 83$, or $l_v + u_v \leq 82$ and both core quantiles hold. For ordinary endpoints, either $v + l_v \geq b + 5$, or $l_v \geq b + 5$ and $v \leq -1$; either $v - u_v \leq S_p - a - 4$, or $u_v \geq a + 5$ and $v \geq S_p + 1$. For each inner rank $i = 1, \ldots, 4$, either $l_v \geq i$ or $v \geq y_{5-i} + 1$, and either $u_v \geq i$ or $v \leq S_p - x_{5-i}$. Finally a potentially split pair has either the necessary condition for its other contributor, or its selected actual member is sole and its two hull losses can be added to the individual-loss budget, with global quantiles 87 and 86.

These are alternatives for actual configurations, whose losses and centers are integral. The model relaxes the coordinates to nonnegative reals. After transport an upper-endpoint ordinary alternative is selected using the enlarged endpoint; the original upper loss still dominates the lost-point count. A necessary positivity condition is never used as a sufficient condition for contribution. Every binary node retains both children; no hypothesis limiting the number of large-loss centers is imposed.

6 Complete exact coverage and reproducibility

The verifier reconstructs the scalar domain, the complete covering-mode grammar, and every row policy before reading the root assignments. It checks the finite base of Theorem 2.1 on every run. A certificate cannot omit the new lemma's finite cases or enable a conditional row merely by naming it. The final endpoint counts are

Quantity	Count
Structural root groups	6,806
Integer root systems	61,422
Coarse / nested structural groups	6,758 / 48
Direct symbolic interval checks	6,809
Extremum-point evaluations	16,611
Integer roots requiring trees	182
Internal binary nodes / tree leaves	493 / 675
Maximum tree depth	10
Final integer leaf systems	61,915
Distinct integer certificate vectors	897

The split counts are 342 core-or-large, 109 ordinary-or-outside, 11 inner-endpoint, and 31 both-or-sole pair splits. The identities $675 = 182 + 493$ and $61915 = 61422 + 493$ are checked.

For $Ax \le b$, $x \ge 0$, the certificate supplies nonnegative integer row weights w with $w^T A \ge 0$ and $w^T b < 0$. Therefore $0 \le w^T Ax \le w^T b < 0$, a strict contradiction. The minimum normalized gap $-w^T b / \sum_i w_i$ is $1/91$. For each side-size interval, the matrix is fixed and the right-hand sides are directly constructed quadratic polynomials. Exact arithmetic checks their weighted maximum at the endpoints and the integers around the vertex. Every parameter is assigned exactly once; missing roots, unknown rows, unproved or repeated splits, and incomplete trees are rejected.

The 19 negative controls include the invalid one-large-loss shortcut, universal core rows for the Z_2 or H_1 families, nested leading rows at cap 171, a stronger false $r - 4$ defect bound, use of the new theorem below its proved threshold, arbitrary reflection in the active estimate, and an unsupported exclusion of cardinality 94. This verifier proves $N \le 94$, not $N \le 93$.

Verification files. The following commands need only Python 3.10+ and its standard library. The endpoint data are `data/upper94_certificates.json`:

```
python -I -S -B verification/nonap_sharp.py
python -I -S -B verification/upper94.py --output .work/upper94.json
python -I -S -B verification/upper94.py --exhaustive
```

Ordinary mode checks symbolic intervals and all tree leaves. Exhaustive mode additionally rebuilds every integer model, compares every row with its symbolic expression, and checks all assigned weights pointwise. The finite lemma checker recomputes all 2,723 cases and their inequalities; its enumeration digest is an audit record, not a substitute for checking.

The optional SciPy search in `experiments/generate_upper94_certificates.py` saves only candidates that pass exact checking. Results, the exceptional-group index, logs and receipts are in `results/2026-10-16/upper94/`. Standard and optimized Python outputs are byte-identical; exhaustive and interval results agree apart from the mode label; regenerated certificates are byte-identical; and upper-95 regression is unchanged. The Windows entry `verify_upper94_windows.bat` runs all eight finite continuations.

A program PASS checks the finite proof data. The unbounded deleted-interval reduction and real-set argument are proved above; they and the cited foundation have not been formalized in a proof assistant.

Next endpoint. At size 94, $d = 86$, $e = 78$, $m = 77$, and the balanced parameters $\Omega = 467$, $G = 162$ allow a positive low-hole cap $467 - 67 - 162 + 28 = 266 \ge 3 \cdot 86$. The positive three-empty family reappears. That endpoint needs new lattice, threshold and coverage checks.

References

[1] Sum-of-Subset research project, *Joint covering losses and the audited bound $N \le 95$*, `reports/upper95_from96_20261016.tex`, 16 October 2026.

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[4] V. F. Lev, *On consecutive subset sums*, Discrete Mathematics **187** (1998), 151–160, Theorem 1. https://doi.org/10.1016/S0012-365X(98)80006-X.